PRACTICAL MACHINE LEARNING

Principal Component Analysis (PCA)

From the curse of dimensionality to reducing 2 features into 1

One dataset • Every calculation • Visible reference tables • scikit-learn notebook

Student-friendly, step-by-step lesson

Two ways to reduce the number of features

The methods answer different questions.

Feature selection

Keep some original columns.

From: X1, X2, X3, X4

To: X1, X3

PCA — feature extraction

Create NEW columns by combining the originals.

From: X1, X2, X3, X4

To: PC1, PC2

PCA does not simply choose the columns with the highest variances.

What exactly is PCA?

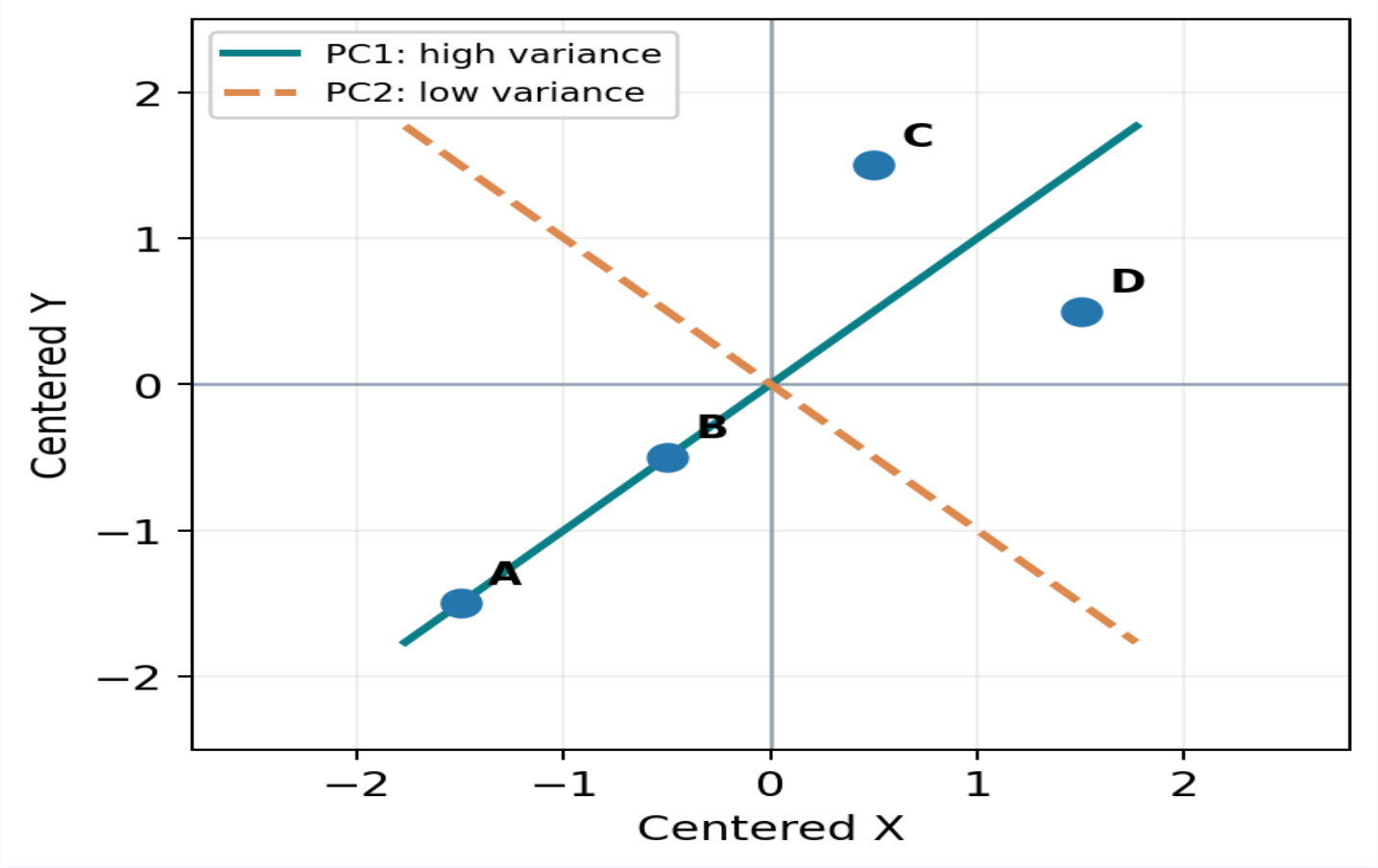
Principal Component Analysis creates new axes that retain as much variance as possible.

Input: 25 columns → **Output: perhaps 2 PCs**

- PC1 = direction containing the greatest variance.
- PC2 = next greatest variance, perpendicular to PC1.
- The number of retained components controls information loss.

Imagine rotating the axes

PCA finds new directions; it does not delete records.



PC1

Direction of greatest variation.

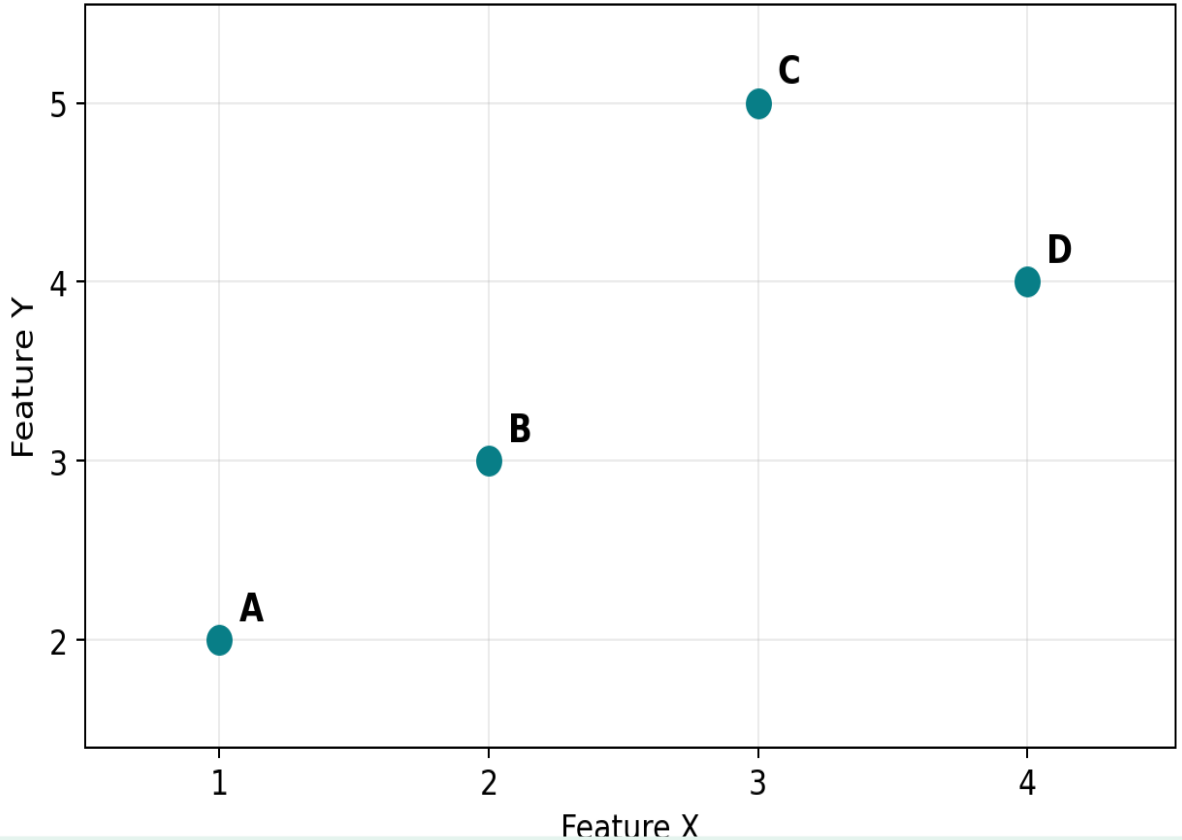
PC2

Perpendicular direction with smaller variation.

Our four-record dataset

We will reduce two original features to one new feature.

Record	X	Y
A	1	2
B	2	3
C	3	5
D	4	4



Keep this dataset in view. All the mathematical steps that follow use these same four rows.

How will PCA transform these numbers?

A complete 2-feature → 1-component calculation.

1. Center X and Y
2. Calculate sample variance and covariance
3. Build the covariance matrix
4. Find new directions and their variances
5. Retain PC1 and project every record

Find the mean of both features

We center each column using its own mean.

Mean of X = (1 + 2 + 3 + 4) / 4 = 2.5

Mean of Y = (2 + 3 + 5 + 4) / 4 = 3.5

We subtract 2.5 from every X value and 3.5 from every Y value.

REFERENCE DATA

ID	X	Y
A	1	2
B	2	3
C	3	5
D	4	4

Original values: A(1,2), B(2,3), C(3,5), D(4,4).

Subtract the mean from each value

Centered X = X − 2.5 | Centered Y = Y − 3.5

Record	X − 2.5	Y − 3.5
A	−1.5	−1.5
B	−0.5	−0.5
C	+0.5	+1.5
D	+1.5	+0.5

Both centered columns now have mean 0. The record-to-record geometry is unchanged.

REFERENCE DATA

ID	X	Y
A	1	2
B	2	3
C	3	5
D	4	4

Centering changes the origin, not the distances between records.

What does variance measure?

Variance measures the average squared deviation from the mean.

$$\text{Sample variance} = \Sigma(\text{value} - \text{mean})^2 / (n - 1)$$

X	X − 2.5	Squared deviation
1	−1.5	2.25
2	−0.5	0.25
3	+0.5	0.25
4	+1.5	2.25
Total		5.00

Var(X) = 5 / 3 = 1.6667

REFERENCE DATA

ID	X	Y
A	1	2
B	2	3
C	3	5
D	4	4

For sample variance, the denominator is n − 1 = 3.

Calculate variance of Y

Exactly the same rule: subtract its mean, square, add, then divide by 3.

Y	Y − 3.5	Squared deviation
2	−1.5	2.25
3	−0.5	0.25
5	+1.5	2.25
4	+0.5	0.25
Total		5.00

Var(Y) = 5 / (4 − 1) = 1.6667

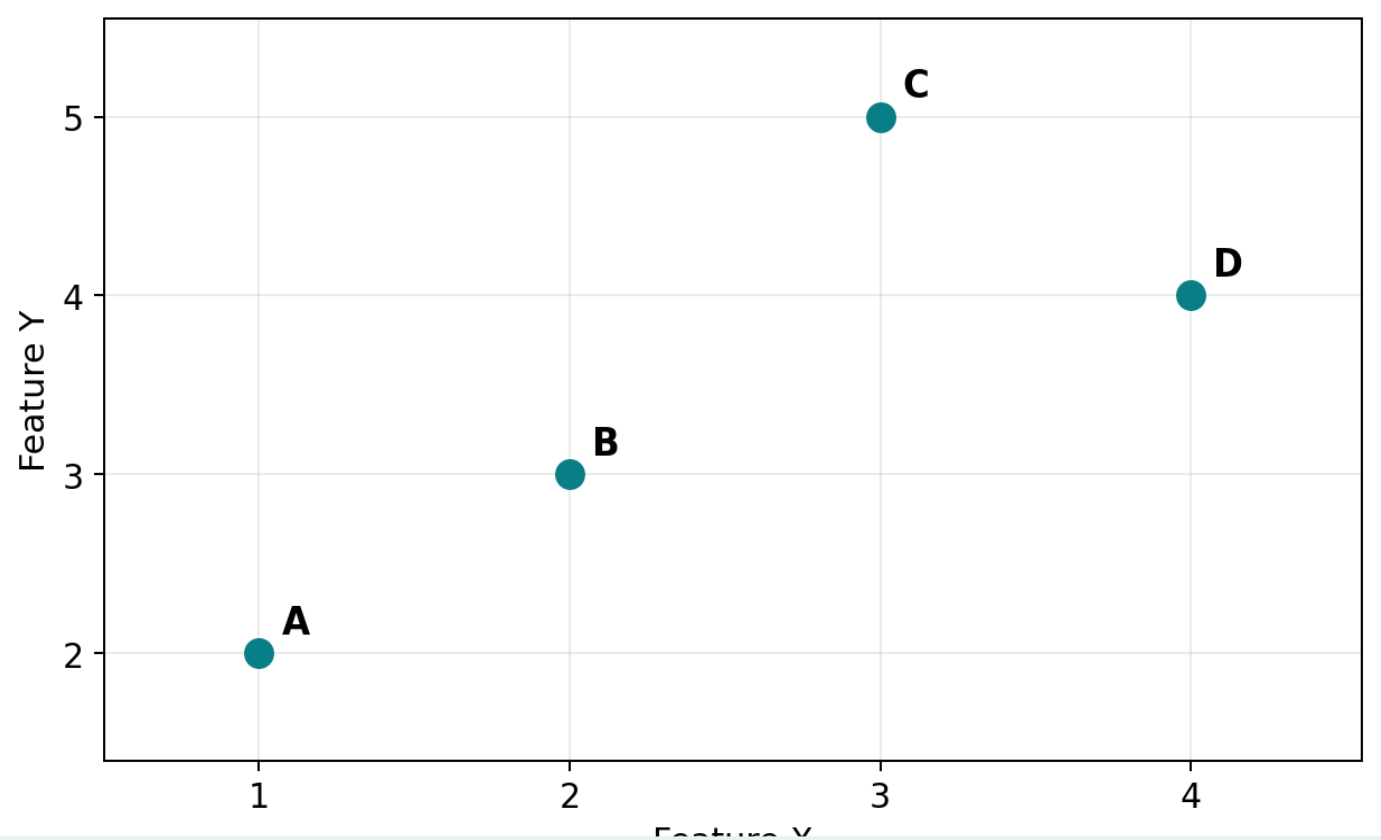
REFERENCE DATA

ID	X	Y
A	1	2
B	2	3
C	3	5
D	4	4

X and Y happen to have equal variance in this dataset.

Why is variance alone not enough?

X and Y both have variance 1.6667 — which direction should PCA choose?



Both features vary equally

But they tend to increase together. Variance of X and Y alone cannot describe this shared movement.

We need COVARIANCE: how do X and Y vary together?

What does covariance mean?

Positive: they rise together. Negative: one rises while the other tends to fall.

Positive covariance

When X is above its mean, Y is often above its mean.

Negative covariance

When X is above its mean, Y is often below its mean.

$$\mathrm{Cov}(X,Y) = \Sigma[(X-\bar{X})(Y-\bar{Y})] / (n-1)$$

REFERENCE DATA

ID	X	Y
A	1	2
B	2	3
C	3	5
D	4	4

Our dataset has positive covariance between X and Y.

Calculate covariance row by row

Multiply each pair of centered values, add, and divide by n − 1 = 3.

Record	X − 2.5	Y − 3.5	Product
A	−1.5	−1.5	2.25
B	−0.5	−0.5	0.25
C	+0.5	+1.5	0.75
D	+1.5	+0.5	0.75
Sum			4.00

Cov(X,Y) = 4 / 3 = 1.3333

REFERENCE DATA

ID	X	Y
A	1	2
B	2	3
C	3	5
D	4	4

Use the same centered X and Y values obtained earlier.

Why does PCA need the covariance matrix?

It needs BOTH individual spread and shared movement to choose the best NEW axis.

Individual variance

How far does X spread? How far does Y spread?

Covariance

When X moves, how does Y tend to move?

Together in a matrix

Calculate variance along ANY direction, not only X or Y.

This is the bridge: covariance matrix → principal directions → new PCA features.

REFERENCE DATA

ID	X	Y
A	1	2
B	2	3
C	3	5
D	4	4

Var(X)=Var(Y)=1.6667; covariance=1.3333.

Arrange these values in the covariance matrix

Diagonal = each feature's variance; off-diagonal = shared variation.

$$C = \begin{bmatrix} 1.6667 & 1.3333 \\ 1.3333 & 1.6667 \end{bmatrix}$$

Var(X) = top-left Var(Y) = bottom-right

Cov(X,Y) = both off-diagonal entries

REFERENCE DATA

ID	X	Y
A	1	2
B	2	3
C	3	5
D	4	4

Sample covariance and sample variance use n−1 = 3.

How do we find the principal directions?

From the covariance matrix, calculate eigenvalues first, then eigenvectors.

START WITH C = [5/3 4/3 ; 4/3 5/3]

1 FIND THE EIGENVALUES — how much variance?

$\det(C - \lambda I) = 0$

$(5/3 - \lambda)^2 - (4/3)^2 = 0 \rightarrow \lambda_1 = 3,\ \lambda_2 = 1/3$

2 FIND THE EIGENVECTORS — which directions?

For $\lambda_1 = 3$: $(C - 3I)v = 0 \rightarrow a = b$

$v_1 = (0.7071, 0.7071)$

$v_2 = (0.7071, -0.7071)$

PC1 uses v_1: its eigenvalue (variance) 3 is the largest.

REFERENCE DATA

ID	X	Y
A	1	2
B	2	3
C	3	5
D	4	4

We already know Var(X) = Var(Y) = 5/3 and Cov(X,Y) = 4/3.

Compare the two diagonal directions

The signs of the weights change the contribution from covariance.

Direction	Unit weights	Variance
Together ↗	(+0.7071, +0.7071)	3.0000
Opposite ↘	(+0.7071, −0.7071)	0.3333
X only	(1, 0)	1.6667
Y only	(0, 1)	1.6667

PCA looks for the direction with the MAXIMUM variance: the “together” direction.

REFERENCE DATA

ID	X	Y
A	1	2
B	2	3
C	3	5
D	4	4

Both directions have length 1; their weights are normalized.

Our covariance matrix gives two components

PC1 has the largest variance. PC2 is perpendicular to PC1.

Component	Eigenvector	Eigenvalue
PC1	(0.7071, 0.7071)	3.0000
PC2	(0.7071, −0.7071)	0.3333

Covariance matrix × eigenvector = eigenvalue × eigenvector

REFERENCE DATA

ID	X	Y
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B	2	3
C	3	5
D	4	4

The direction vector can change sign without changing the PCA solution.

How much information does each component keep?

Explained variance ratio = component eigenvalue / sum of eigenvalues.

PC1: 3 / 3.3333 = 90%

PC2: 0.3333 / 3.3333 = 10%



We can keep just PC1 and retain 90% of the ORIGINAL variance.

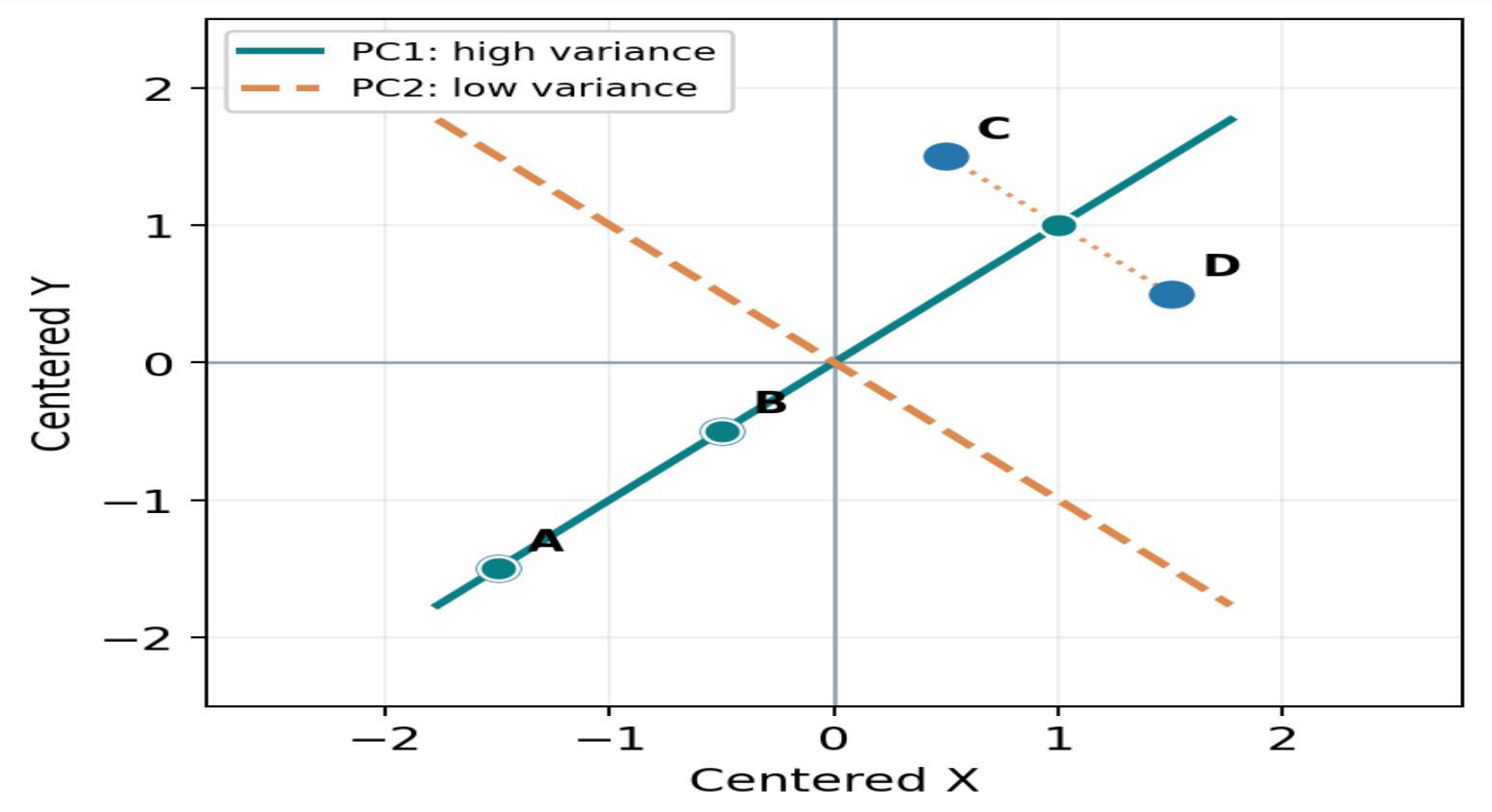
REFERENCE DATA

ID	X	Y
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C	3	5
D	4	4

The total sample variance = 3.0000 + 0.3333 = 3.3333.

What does projection mean?

Now use our calculated PC1 eigenvector (0.7071, 0.7071) to project each CENTERED row.



Each 2D point becomes a 1D coordinate: its position along the new PC1 axis.

REFERENCE DATA

ID	X	Y
A	1	2
B	2	3
C	3	5
D	4	4

PC1 direction = (0.7071, 0.7071).

Calculate the new PC1 value for record A

The new coordinate uses BOTH original features.

$$PC1 = 0.7071 \times \text{Centered X} + 0.7071 \times \text{Centered Y}$$

$$A = 0.7071 \times (-1.5) + 0.7071 \times (-1.5)$$

$$PC1(A) = -2.1213$$

Record	X − 2.5	Y − 3.5
A	−1.5	−1.5
B	−0.5	−0.5
C	+0.5	+1.5
D	+1.5	+0.5

This is why PCA is not column selection: the new value mixes X and Y.

Transform all four records

One new number per row replaces two original feature values.

Record	Centered X	Centered Y	PC1 score
A	−1.5	−1.5	−2.1213
B	−0.5	−0.5	−0.7071
C	+0.5	+1.5	+1.4142
D	+1.5	+0.5	+1.4142

Before: [X, Y] (2 features) → After: [PC1] (1 feature).

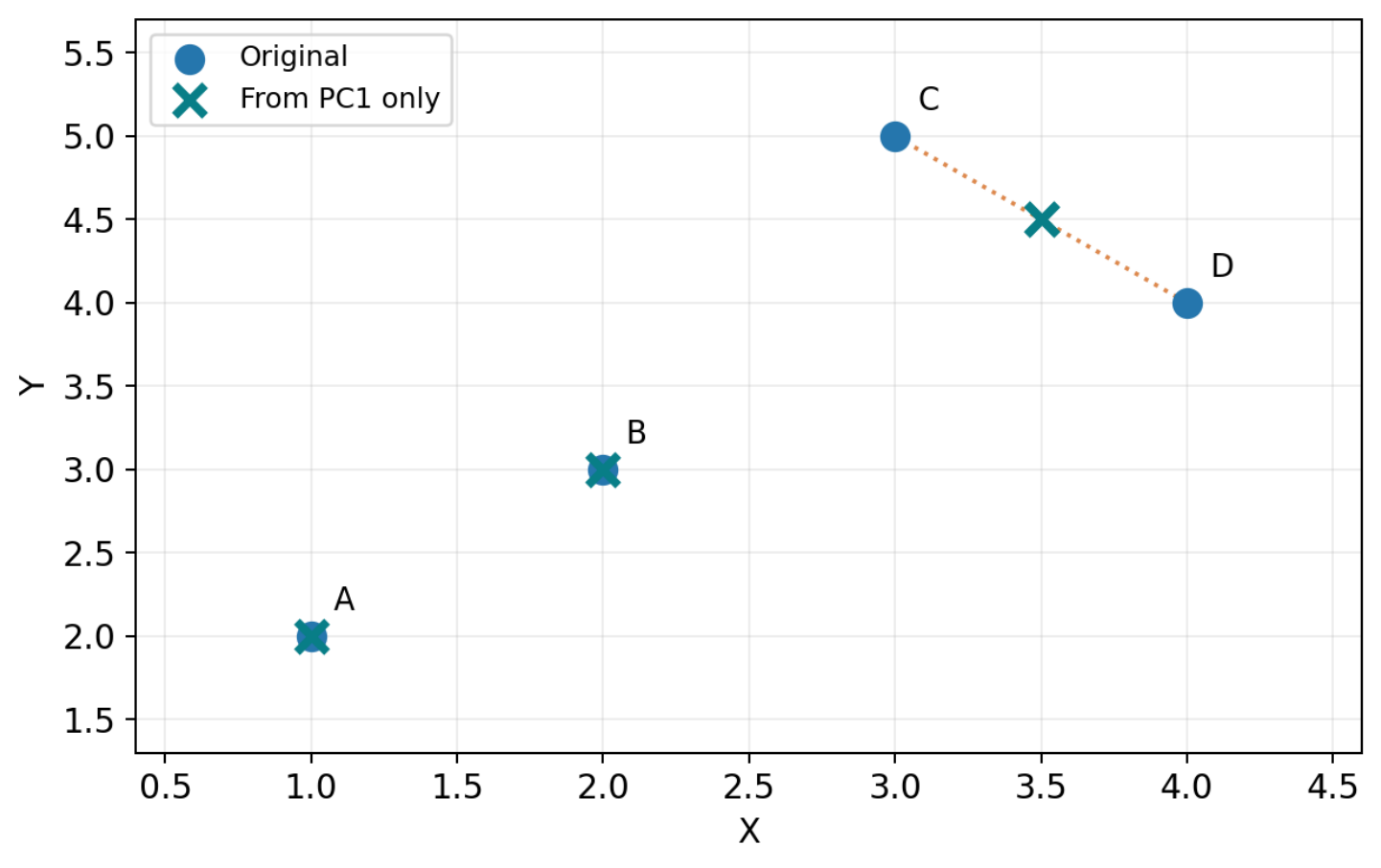
REFERENCE DATA

ID	X	Y
A	1	2
B	2	3
C	3	5
D	4	4

C and D get the same PC1 score because they differ only along PC2.

We have reduced 2 features to 1

PC1 retains 90% of total variance. The remaining 10% is discarded.



C and D overlap on PC1

Their reconstructed coordinates both become (3.5, 4.5).

Different original rows can share a PC1 score.

What does 90% explained variance NOT mean?

PCA is unsupervised: it does not use the target label when finding components.

- 90% explained variance DOES mean: 90% of input feature variance is retained.
- It does NOT mean: classification accuracy is 90%.
- It does NOT guarantee: 90% of information useful for predicting y is retained.

Always evaluate the downstream model — explained variance and model accuracy are not the same.

Do we need standardization before PCA?

Often yes, when features have different units or very different scales.

Feature	Typical values	Without scaling
House size (sq. ft.)	600 – 3,000	Large numerical spread
Number of rooms	1 – 6	Small numerical spread
Age (years)	20 – 80	Different scale

StandardScaler: z = (x − training mean) / training std

The four-record manual example uses raw values because X and Y have equal variances and the same scale.

Reproduce the manual PCA example in scikit-learn

The library computes the same component scores; eigenvector signs may be reversed.

```
df = pd.DataFrame({'X': [1, 2, 3, 4], 'Y': [2, 3, 5, 4]})
print(df.cov())                    # covariance matrix
pca = PCA(n_components=1)
X_pca = pca.fit_transform(df)  # learn PC1, then project
print(pca.components_)         # weights/direction
print(pca.explained_variance_) # variance along PC1
print(pca.explained_variance_ratio_)  # [0.90]
```

A real dataset: 30 features → fewer PCs

PCA can be used before classification, but the target must not influence PCA fitting.

1 • Split

X_train and X_test

→

2 • Scale

Learn means/std only from X_train

→

3 • PCA

Learn principal components only from training data

→

4 • Classify

Fit Logistic Regression on PC scores

Using a Pipeline keeps the training and prediction transformations in the correct order.

What happens during model.fit(X_train, y_train)?

Each stage learns parameters from training data only.

StandardScaler.fit_transform

Learn mean/std and scale training X

↓

PCA.fit_transform

Learn principal axes and project training X

↓

LogisticRegression.fit

Learn classification model from PC scores

What happens during model.predict(X_test)?

The test data is transformed automatically using previously learned parameters.

StandardScaler.transform

Use TRAIN means/std; do not fit again

↓

PCA.transform

Use TRAIN PC directions; do not fit again

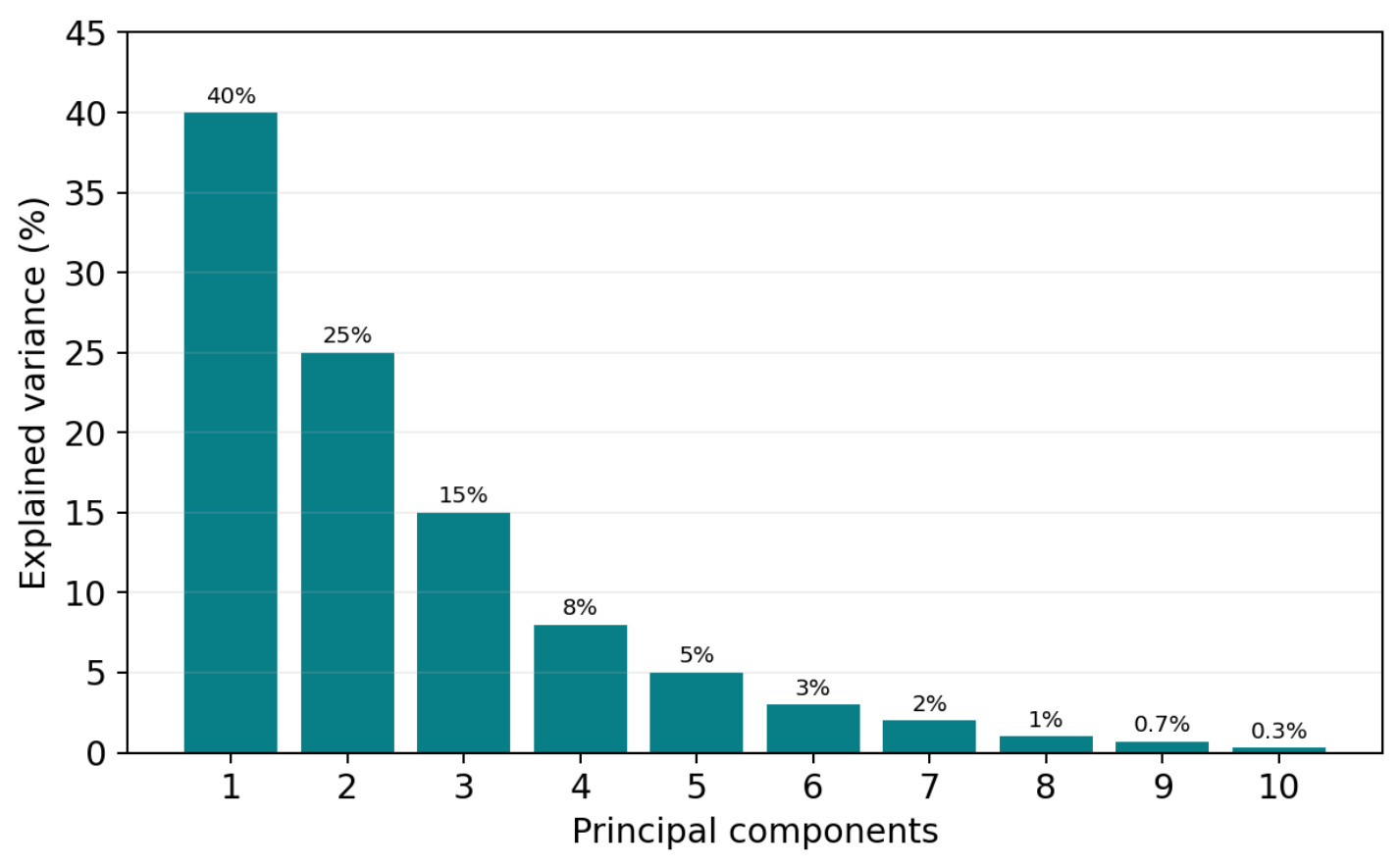
↓

LogisticRegression.predict

Make predictions from transformed test rows

How many components should we keep?

This is an illustrative scree plot for 10 features (not our four-record dataset).



Cumulative variance

PC1–PC3: 80%

PC1–PC5: 93%

n_components=0.95

Chooses enough PCs to retain at least 95% variance.

Where PCA helps — and where it may not

Component retention should be chosen using validation performance, not variance alone.

Consider PCA when...	Think carefully when...
Many numeric features with overlap	Original feature meaning must be preserved
Distance-based algorithms struggle	Predictive signal is low-variance
Need 2D or 3D visualization	Tree-based models already work well
Want a compact feature representation	Features are categorical/unscaled

For KNN / Logistic Regression, compare model scores with and without PCA on the same split or CV folds.

The full PCA story — one more time

Every mathematical step has a clear purpose.

Curse of dimensionality	Motivation for reduction
Center and (when needed) scale	Put features on a fair basis
Variance + covariance	How features vary alone and together
Covariance matrix	Summarize relationships across feature directions
Eigenvectors / eigenvalues	Find directions and corresponding variances
Explained variance + projection	Choose PCs and produce new feature values

Key idea: PCA creates new axes that capture as much variance as possible with fewer dimensions.